PREVIOUSLY

We presented a disk-oriented architecture where the DBMS assumes that the primary storage location of the database is on non-volatile disk.

We then discussed a page-oriented storage scheme for organizing tuples across heap files.

PAGE LAYOUT

For any page storage architecture, we now need to decide how to organize the data inside of the page.

→ We are still assuming that we are only storing tuples in a row-oriented storage model.

→ We will also assume that each tuple fits in a single page.

Approach #1: Tuple-oriented Storage

Approach #2: Index-organized Storage ← Today

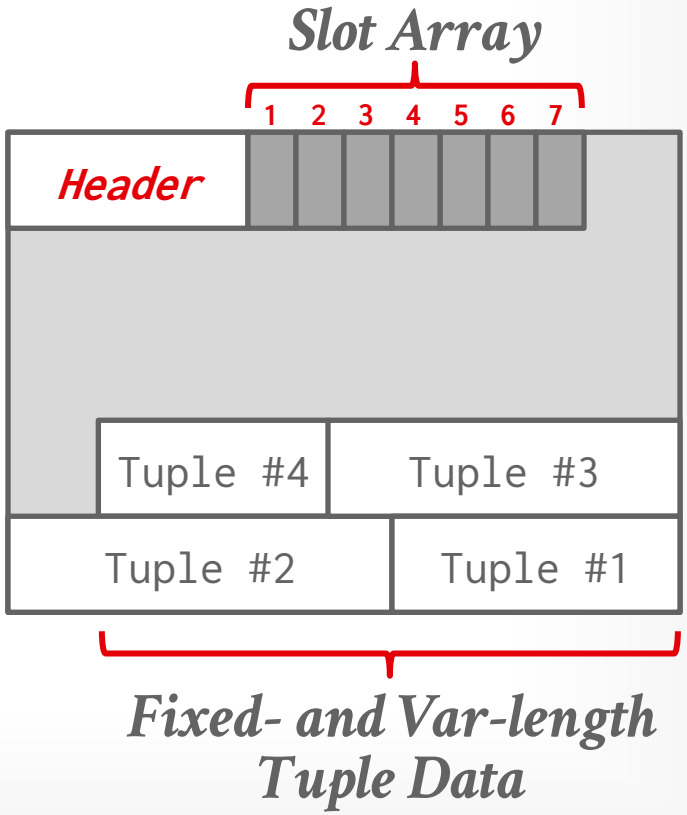
Approach #3: Log-structured Storage

SLOTTED PAGES

The most common page layout scheme is called slotted pages.

The slot array maps "slots" to the tuples' starting position offsets.

The header keeps track of:
→ The # of used slots
→ The offset of the starting location of the last slot used.



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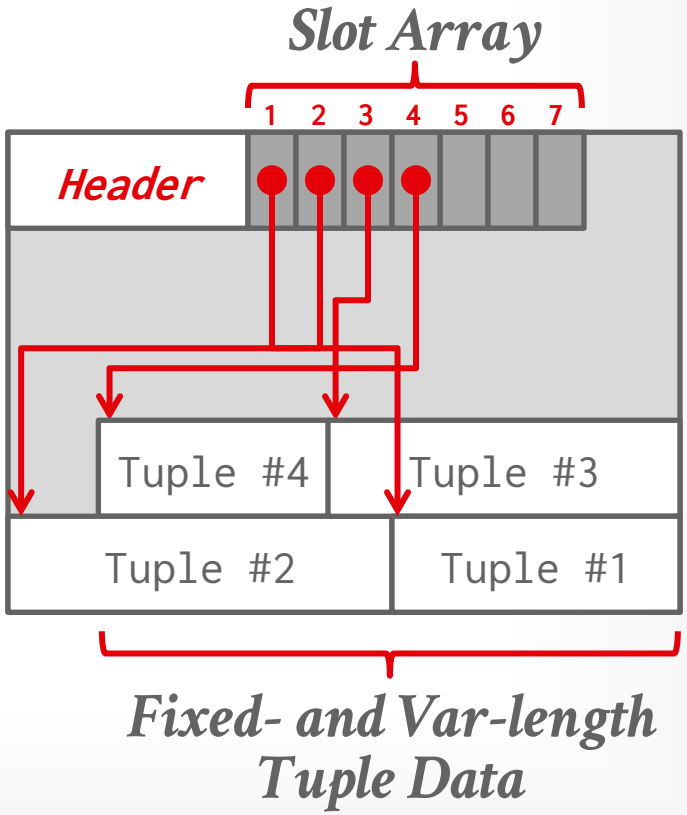

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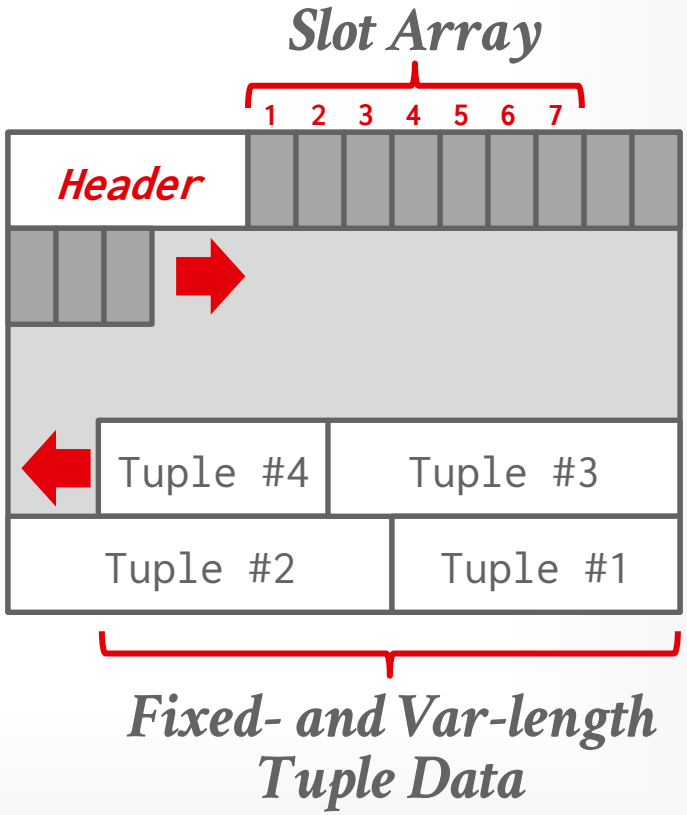

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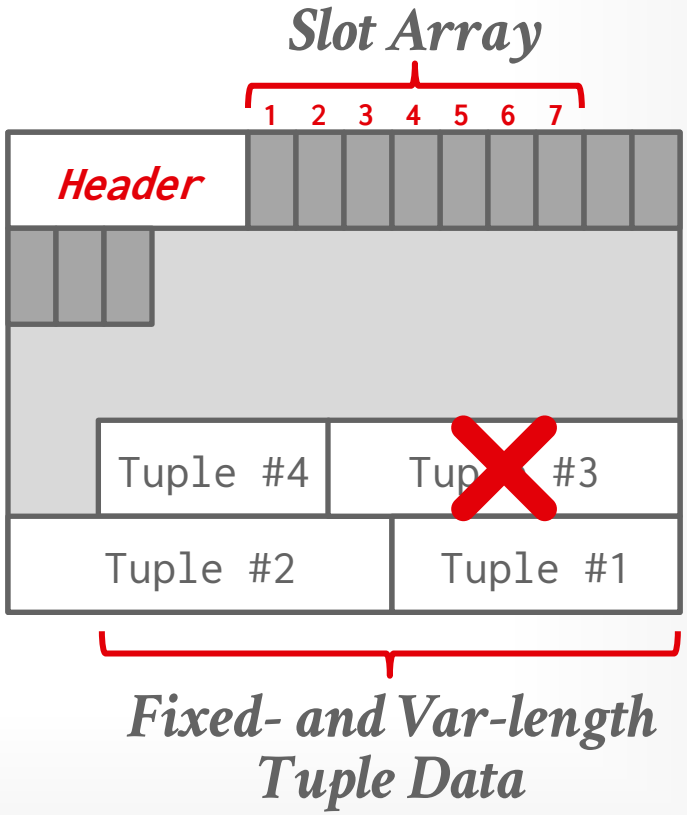

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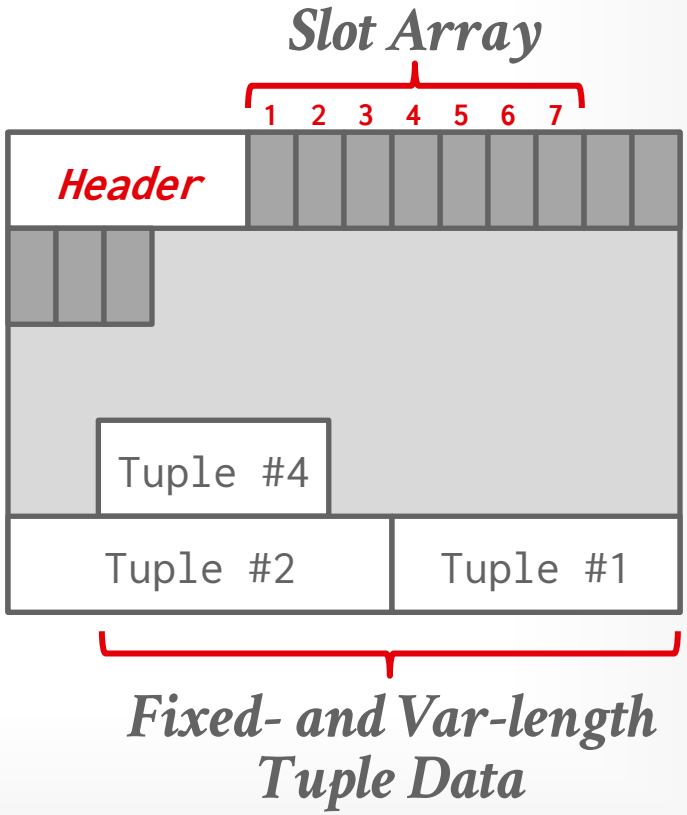

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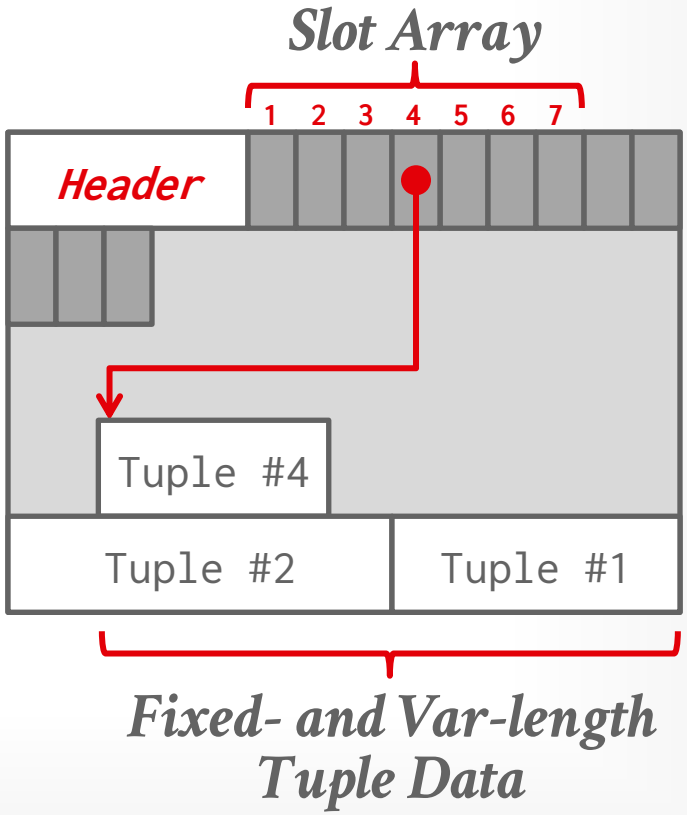

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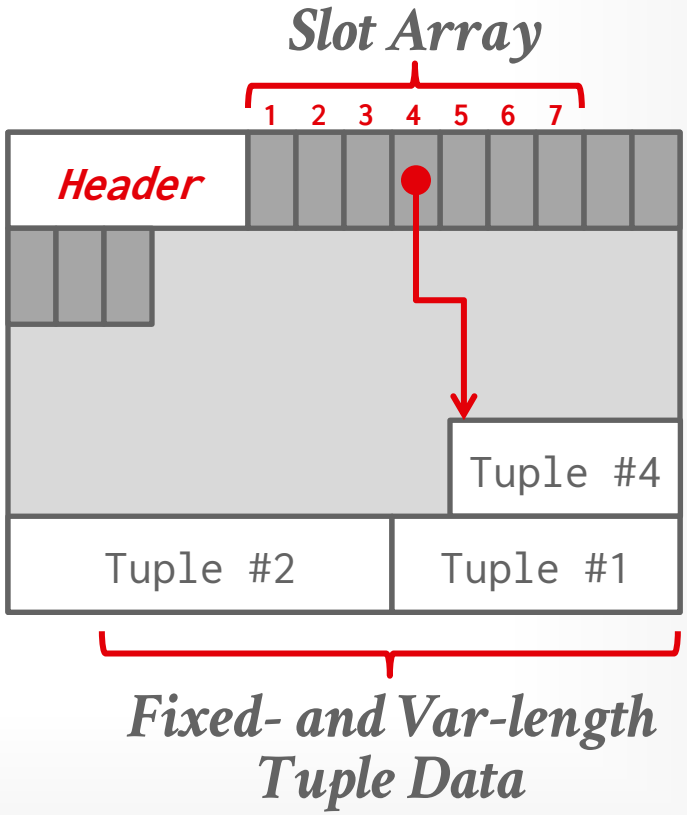

RECORD IDS

The DBMS assigns each logical tuple a unique **record identifier** that represents its physical location in the database.

→ Example: File Id, Page Id, Slot #
→ Most DBMSs do not store ids in tuple.
→ SQLite uses **ROWID** as the true primary key and stores them as a hidden attribute.

Applications should never rely on these IDs to mean anything.

Record Id Sizes

INGRES	TID	*4-bytes*
PostgreSQL	CTID	*6-bytes*
SQLite	ROWID	*8-bytes*
Microsoft SQL Server	%%physloc%%	*8-bytes*
Firebird	RDB$DB_KEY	*8-bytes*
ORACLE	ROWID	*10-bytes*

TUPLE-ORIENTED STORAGE: READS

Get an existing tuple using its record id:

→ Check page directory to find location of page.
→ Retrieve the page from disk (if not in memory).
→ Find offset in page using slot array.

The DBMS relies on indexes to find individual tuples because the tables are inherently unsorted.

But what if the DBMS could keep tuples sorted automatically using an index?

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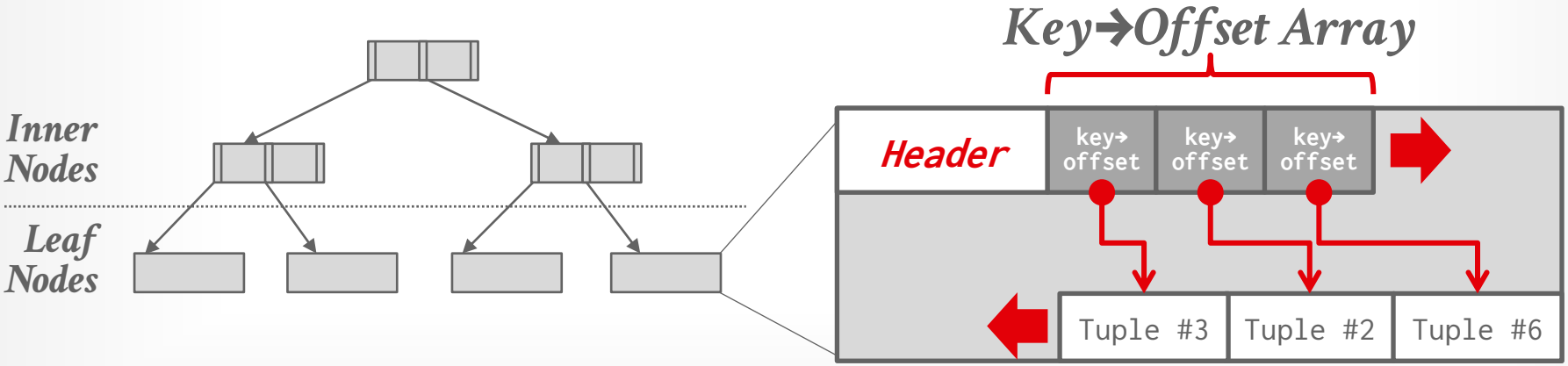
But what if the DBMS could keep tuples sorted automatically using an index?

INDEX-ORGANIZED STORAGE

DBMS stores a table's tuples as the value of an index data structure.

→ Still use a page layout that looks like a slotted page.

→ Tuples are typically sorted in page based on key.



TUPLE-ORIENTED STORAGE: WRITES

Insert a new tuple:

→ Check page directory to find a page with a free slot.
→ Retrieve the page from disk (if not in memory).
→ Check slot array to find empty space in page that will fit.

Update an existing tuple using its record id:

→ Check page directory to find location of page.
→ Retrieve the page from disk (if not in memory).
→ Find offset in page using slot array.
→ If new data fits, overwrite existing data.
Otherwise, mark existing tuple as deleted and insert new version in a different page.

TUPLE-ORIENTED STORAGE

Problem #1: Fragmentation

→ Pages are not fully utilized (unusable space, empty slots).

Problem #2: Useless Disk I/O

→ DBMS must fetch entire page to update one tuple.

Problem #3: Random Disk I/O

→ Worse case scenario when updating multiple tuples is that each tuple is on a separate page.

What if the DBMS <u>cannot</u> overwrite data in pages and could only create new pages?

→ E
xamples: HDFS, Google Colossus, S3 Express

LOG-STRUCTURED STORAGE

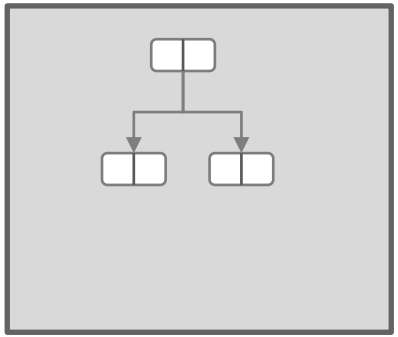
Instead of storing tuples in pages and updating the in-place, the DBMS maintains a log that records changes to tuples.

→ Each log entry represents a tuple **PUT**/**DELETE** operation.
→ Originally proposed as log-structure merge trees (LSM Trees) in 1996.

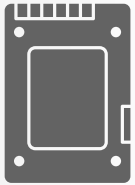
The DBMS applies changes to an in-memory data structure (***MemTable***) and then writes out the changes sequentially to disk as sorted-string tables (***SSTables***).

LOG-STRUCTURED STORAGE

PUT (key101,a_1)



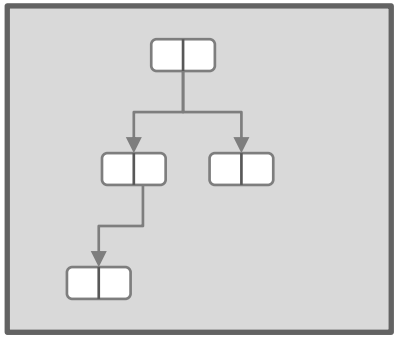
Memory

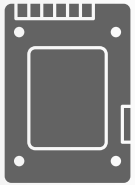


Disk

LOG-STRUCTURED STORAGE

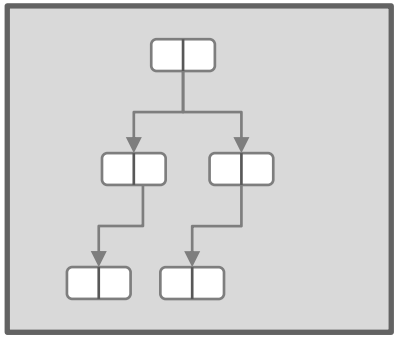
PUT (key101,a_1)

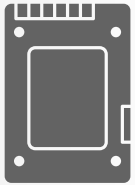




LOG-STRUCTURED STORAGE

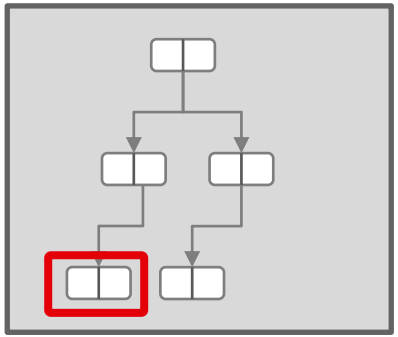
PUT (key102,b_1)

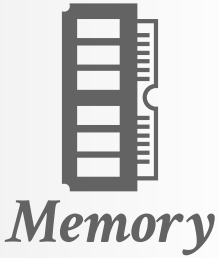


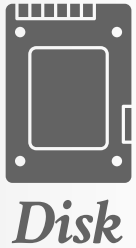


LOG-STRUCTURED STORAGE

PUT (key101,a_2)

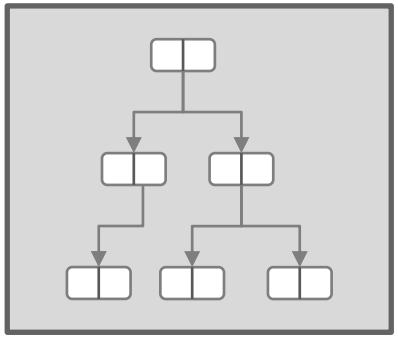


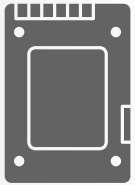




LOG-STRUCTURED STORAGE

PUT (key103,c_1)





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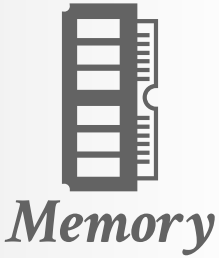

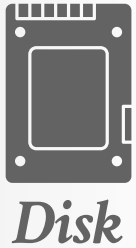

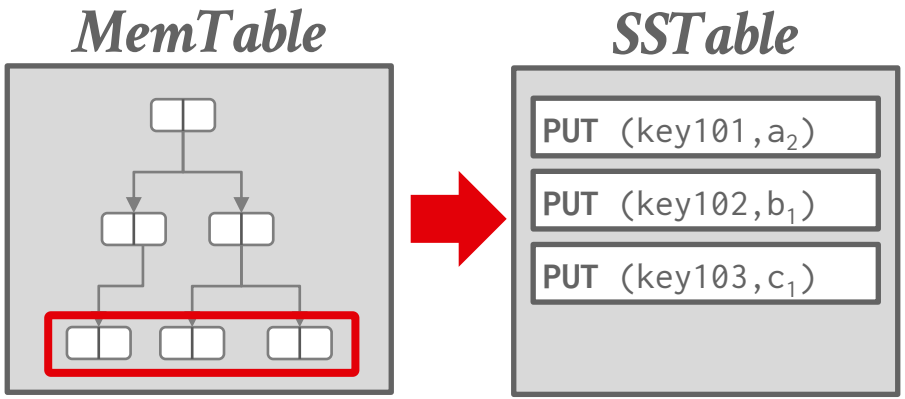

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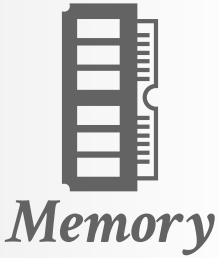

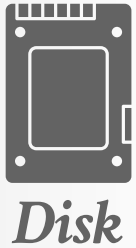

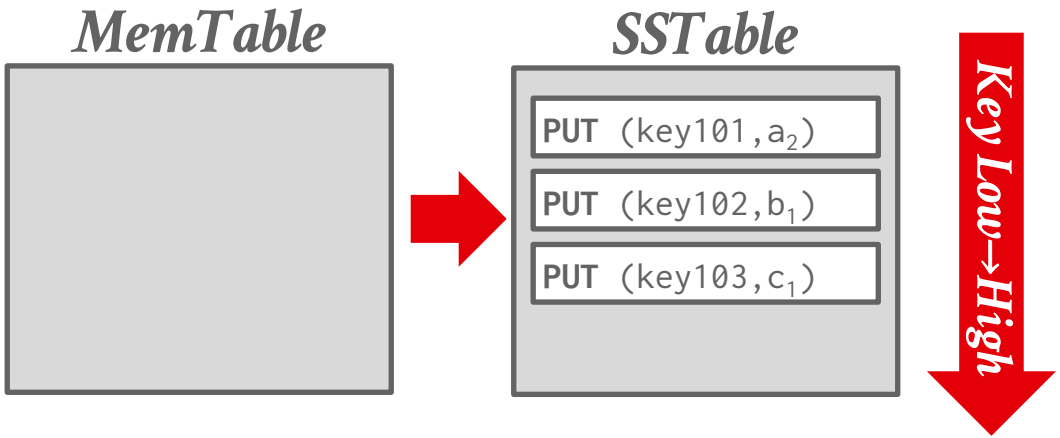

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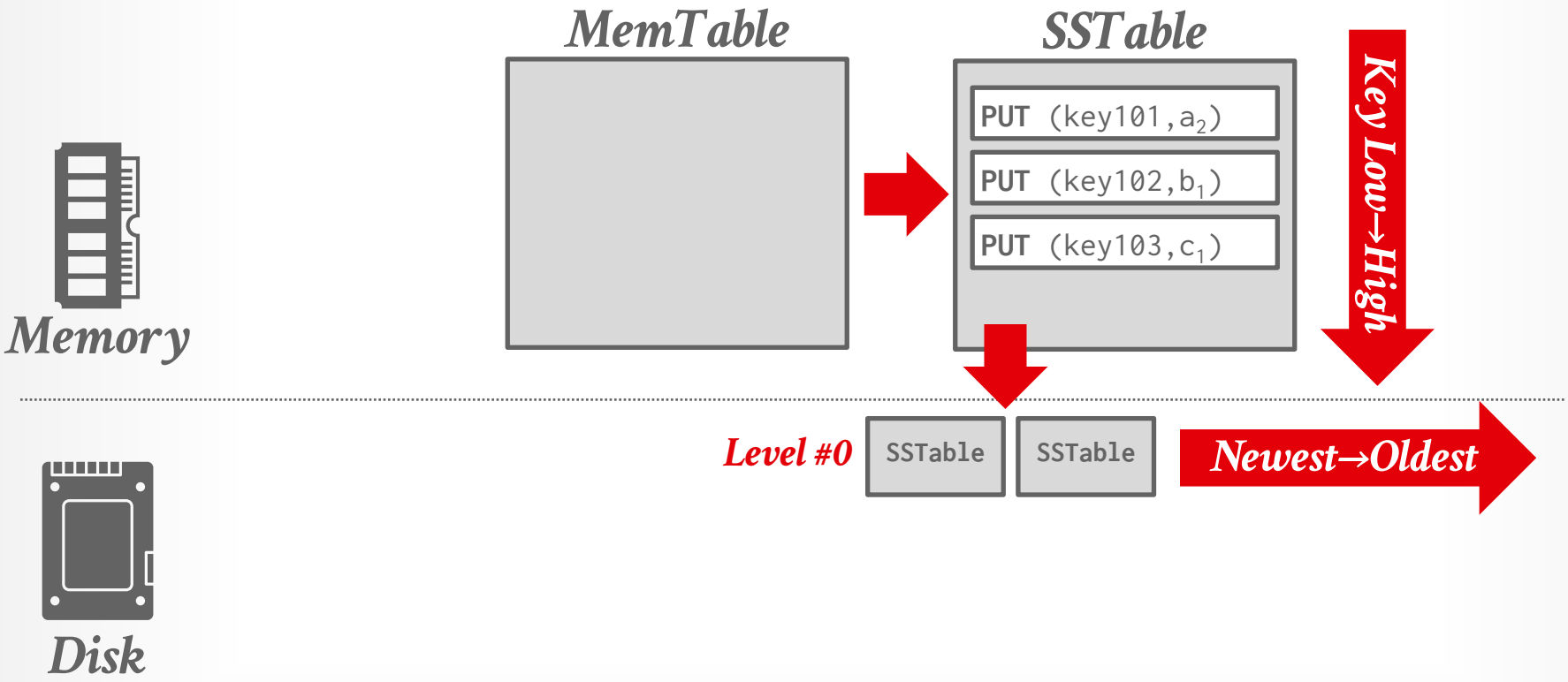

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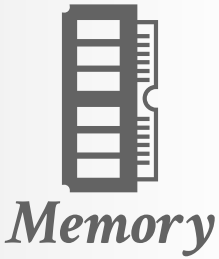

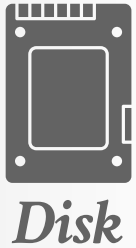

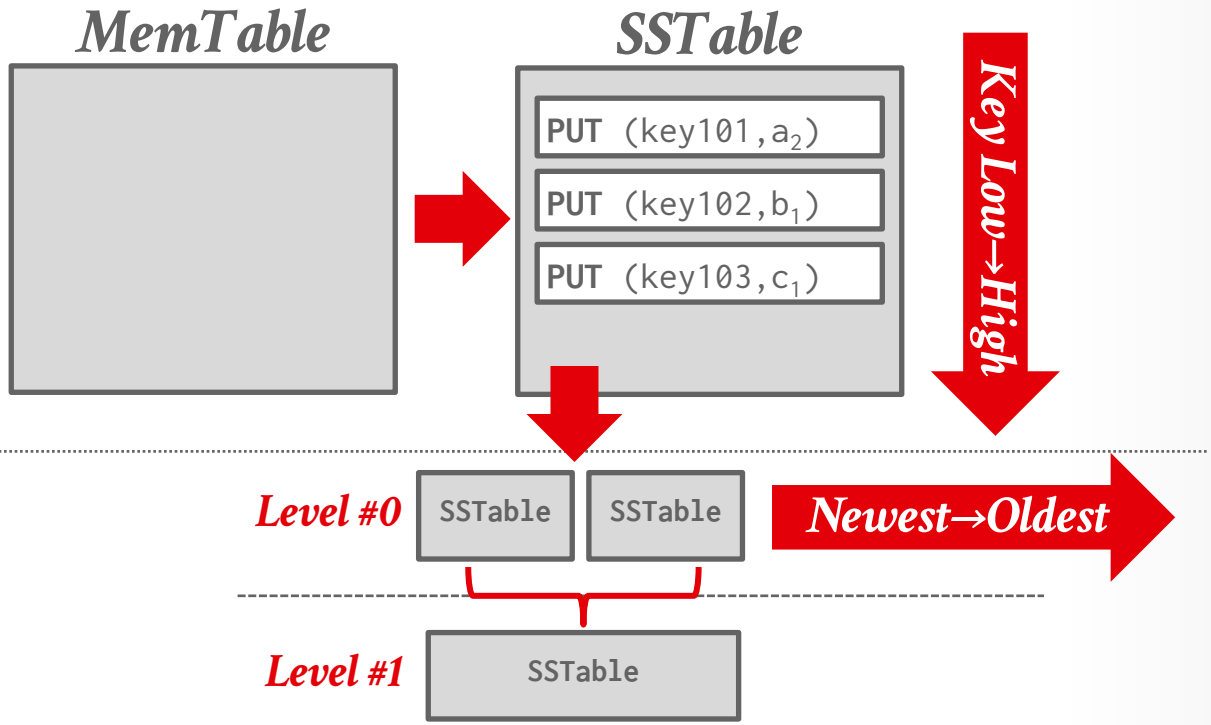

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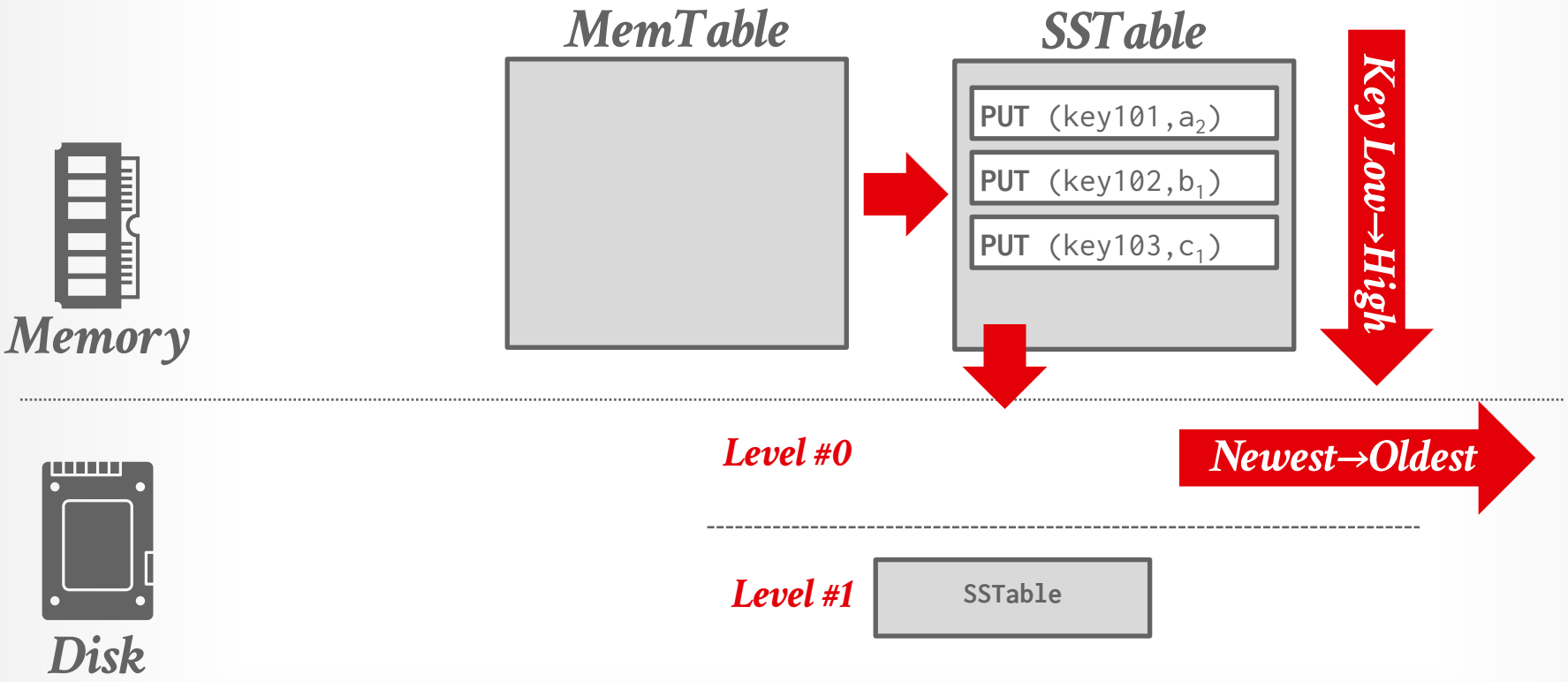

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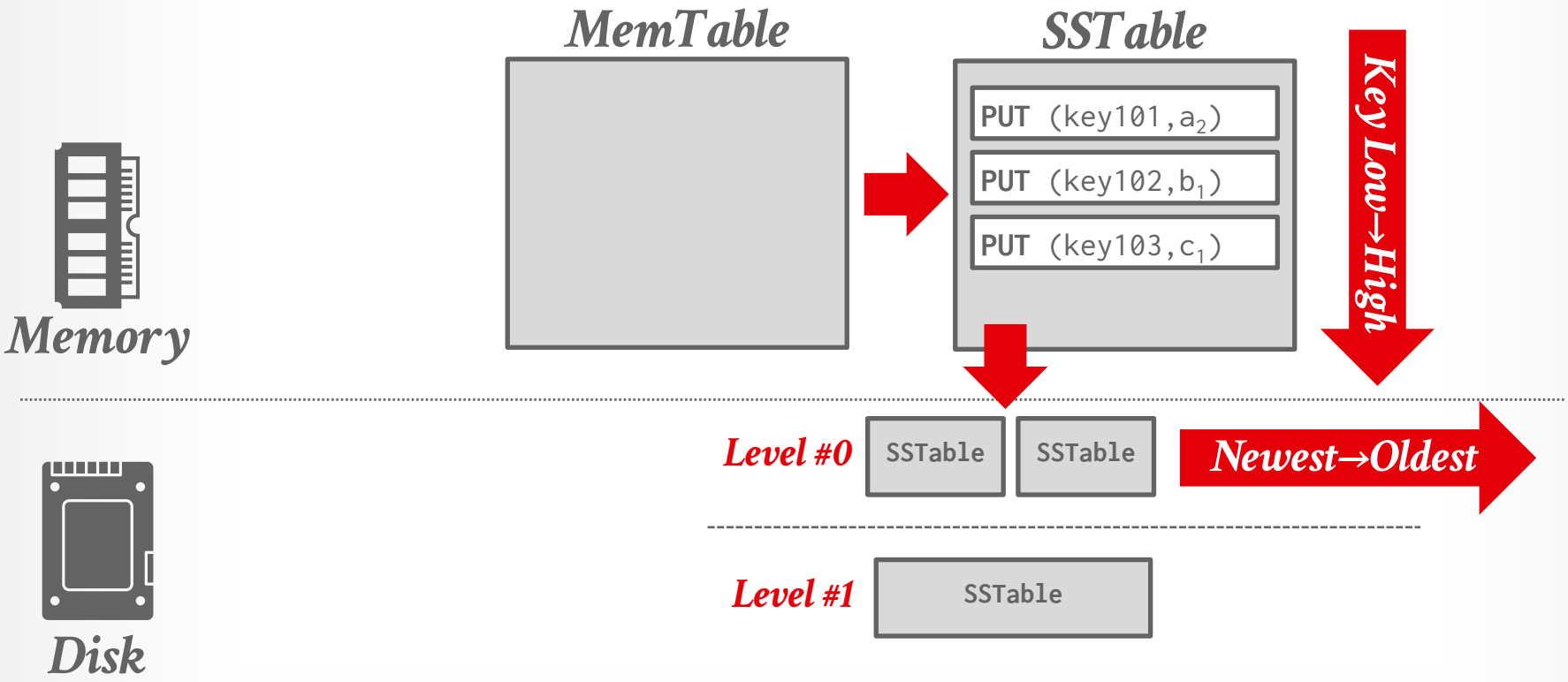

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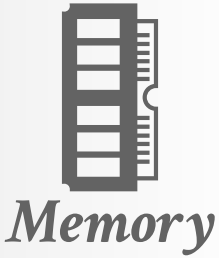

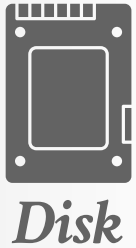

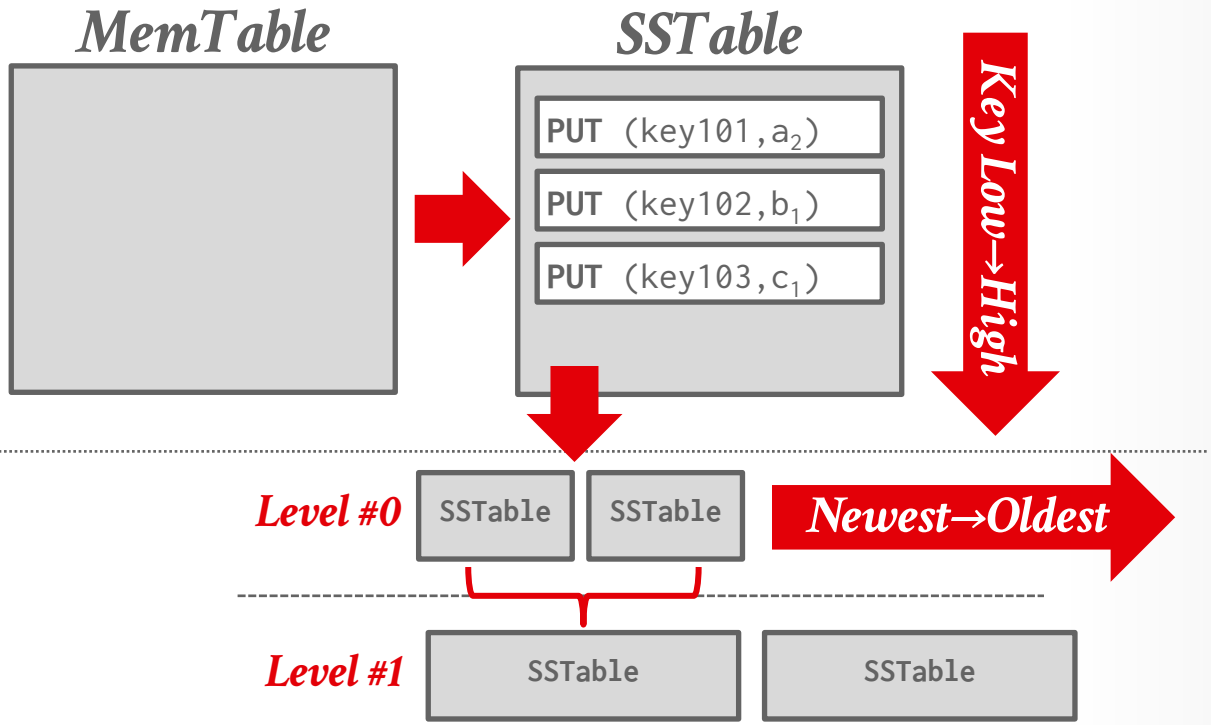

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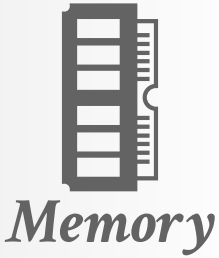

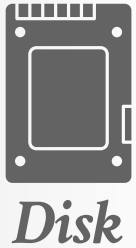

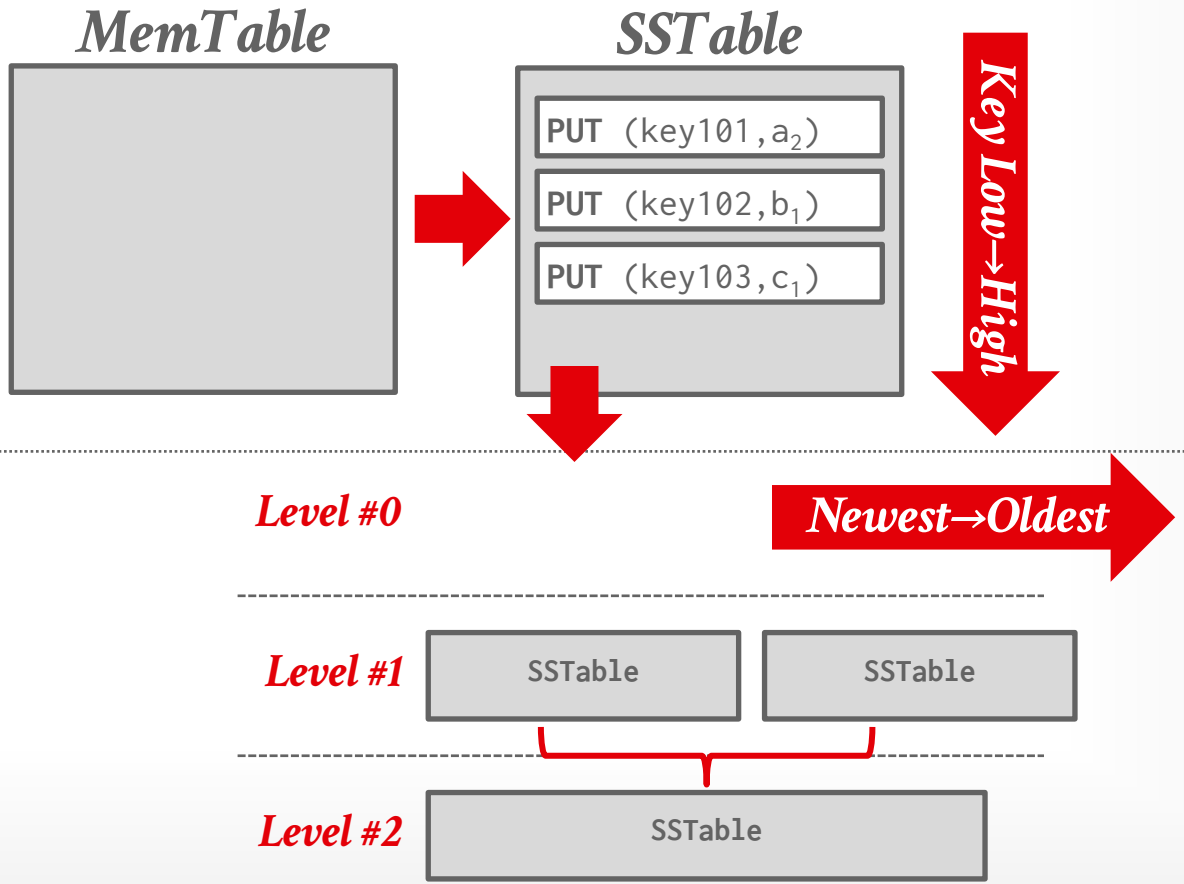

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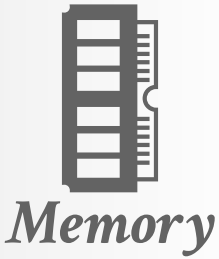
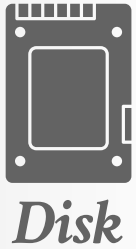




MemTable

SSTable

PUT (key101,a_2)

PUT (key102,b_1)

PUT (key103,c_1)

Key Low→High

Level #0

Newest→Oldest

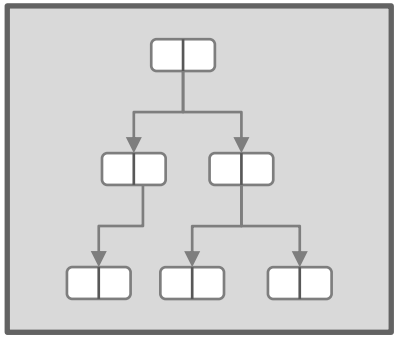
Level #1

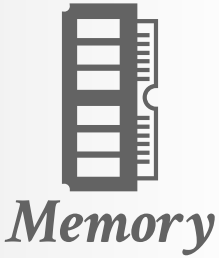
Level #2

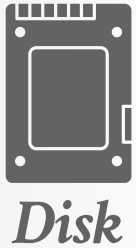
SSTable

LOG-STRUCTURED STORAGE

GET (key101)





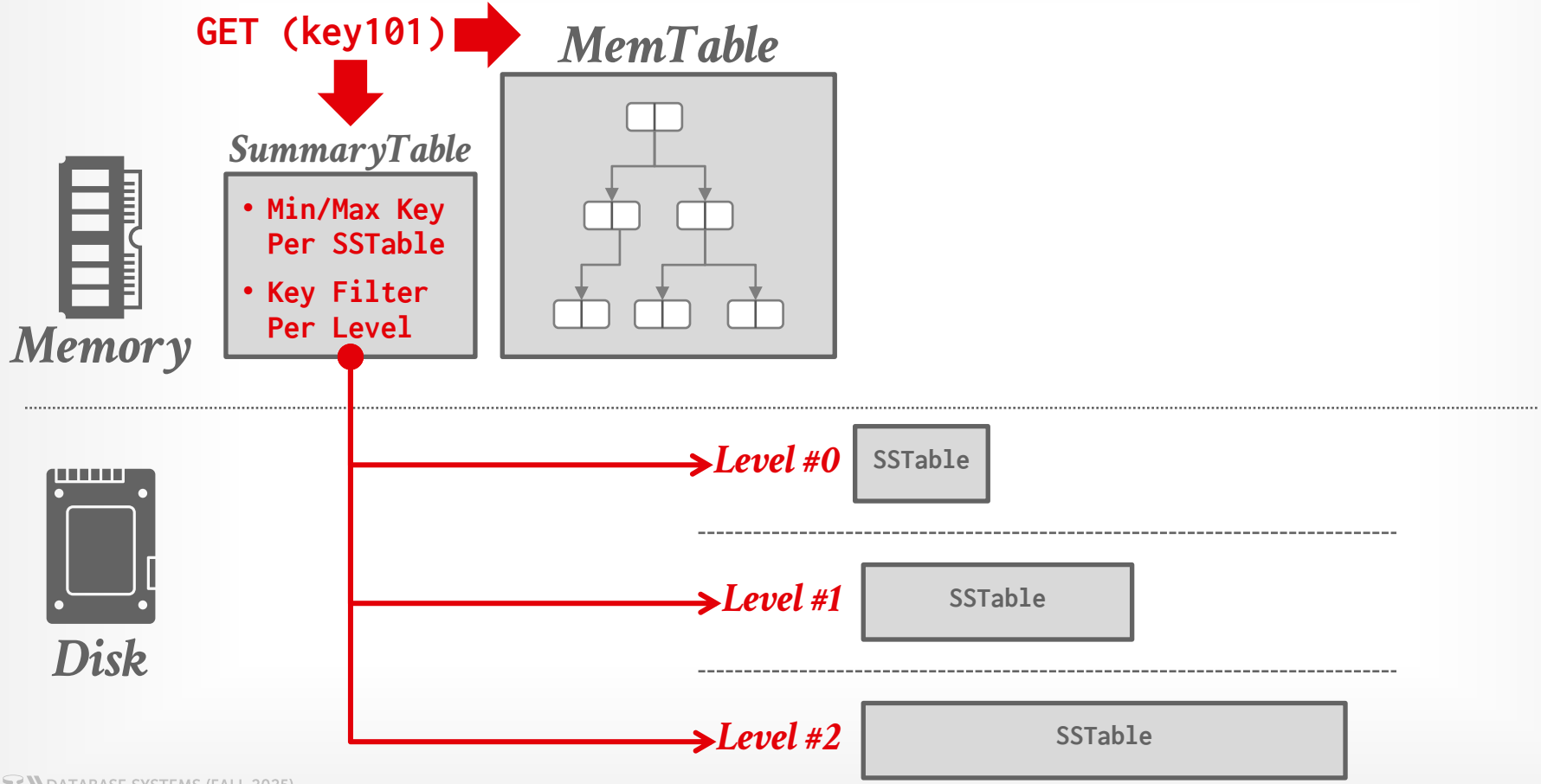


Level #0 SSTable

Level #1 SSTable

Level #2 SSTable

LOG-STRUCTURED STORAGE

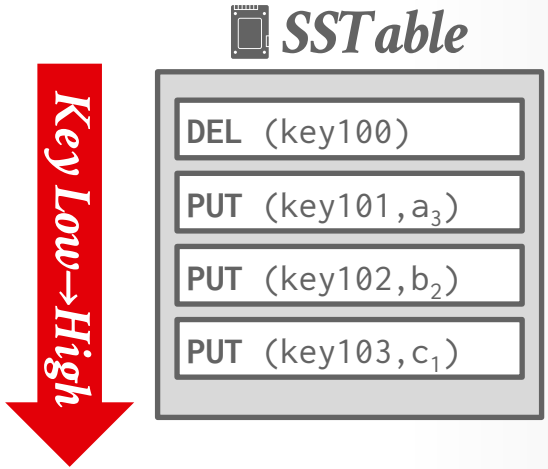


LOG-STRUCTURED STORAGE

Key-value storage that appends log records on disk to represent changes to tuples (**PUT**, **DELETE**).

→ Each log record must contain the tuple's unique identifier.
→ Put records contain the tuple contents.
→ Deletes marks the tuple as deleted.

As the application makes changes to the database, the DBMS appends log records to the end of the file without checking previous log records.



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Periodically compact data files to reduce wasted space and speed up reads.

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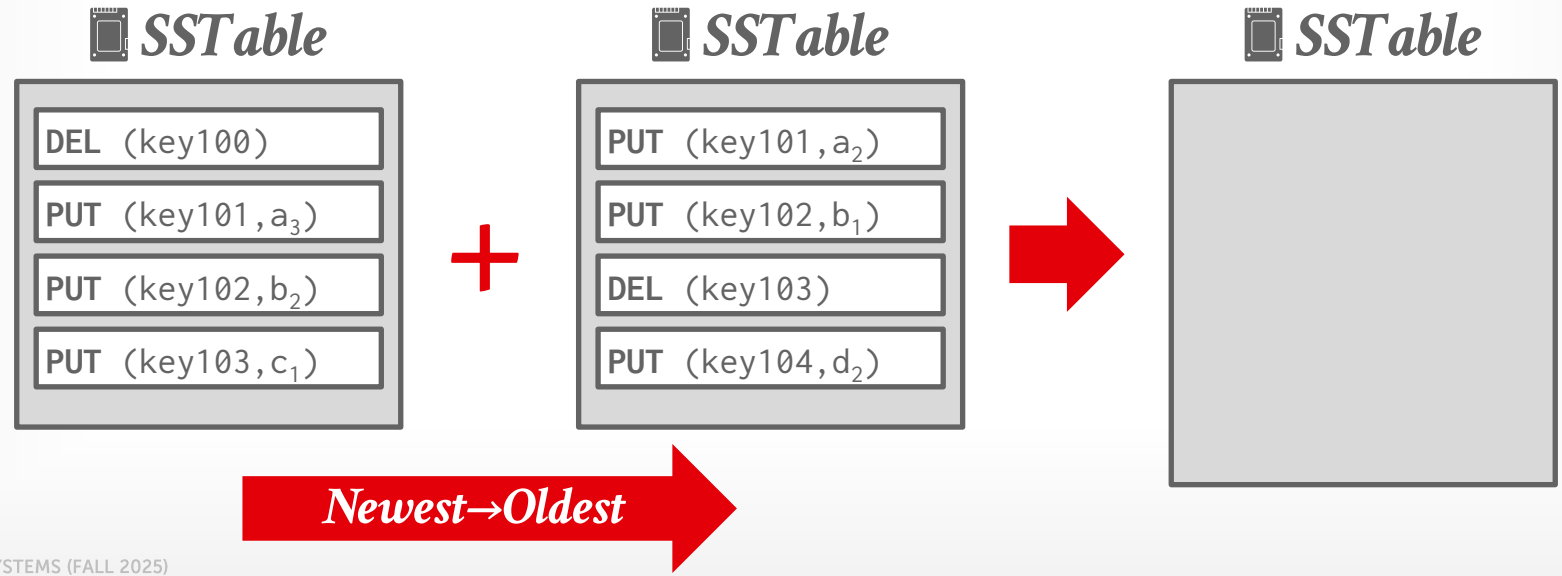

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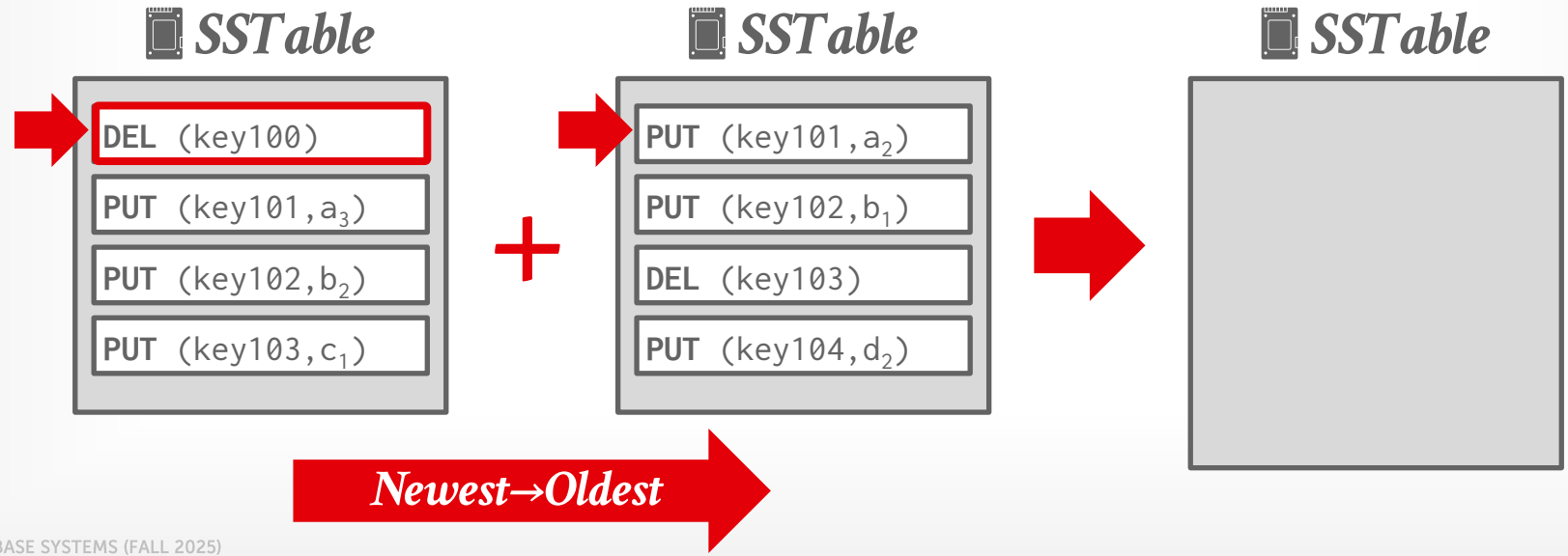

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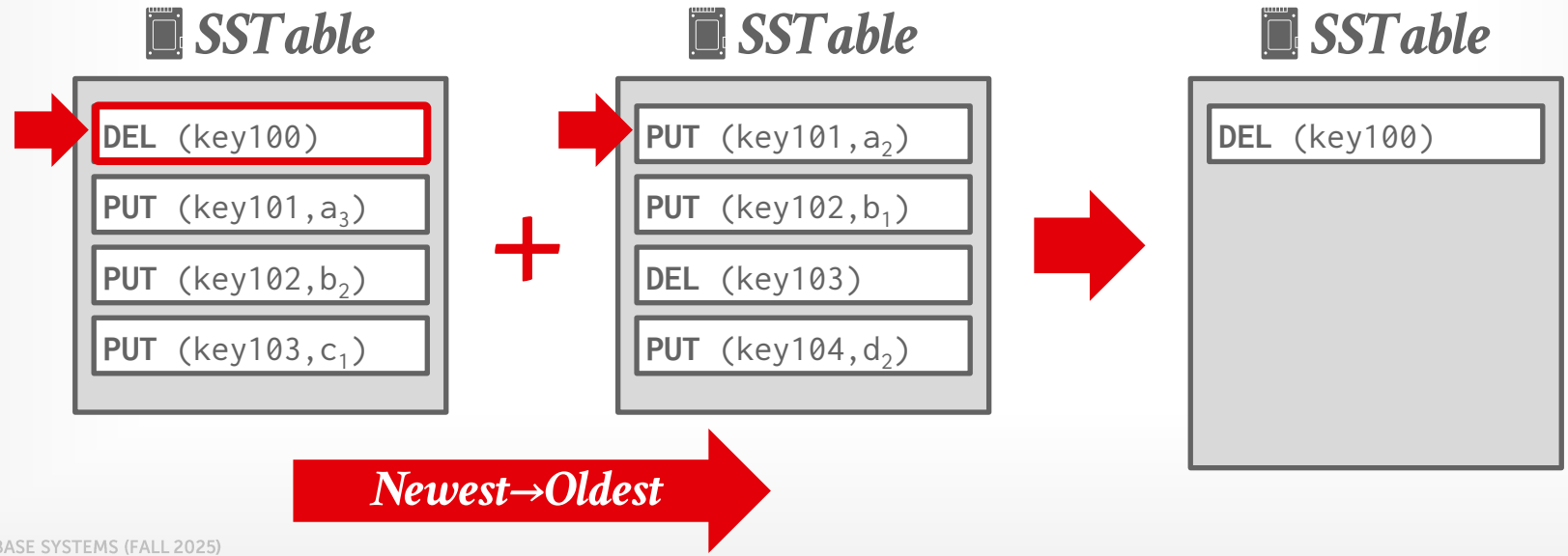

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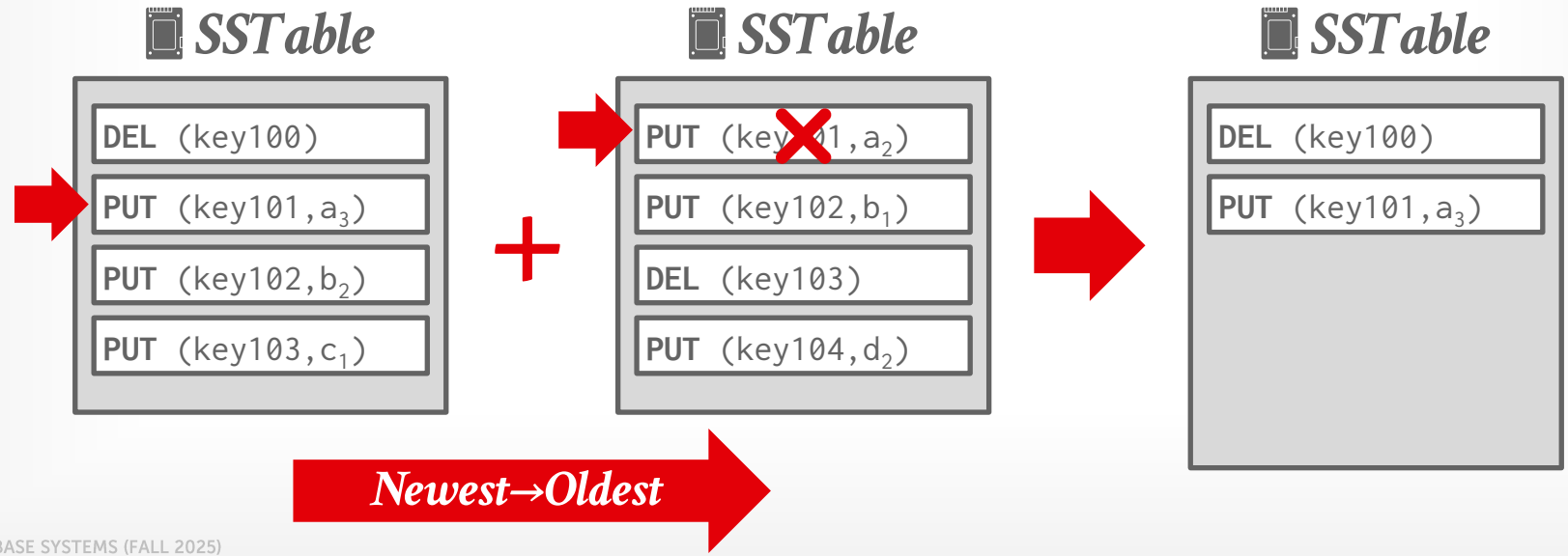

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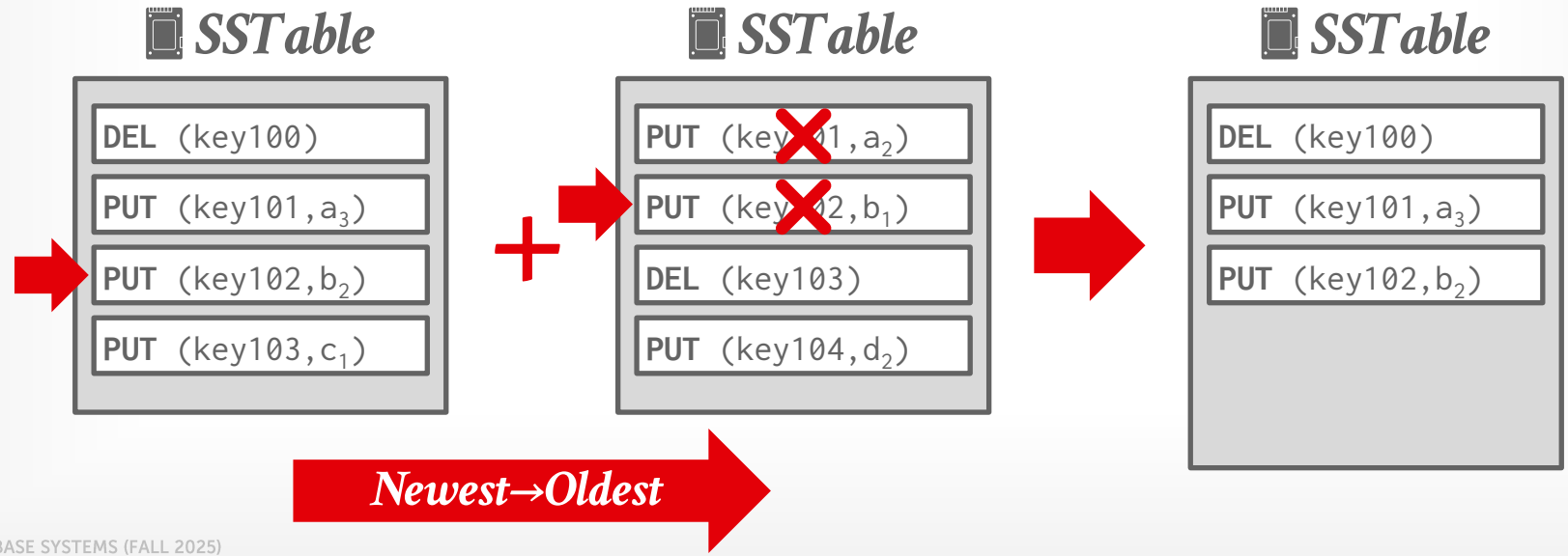

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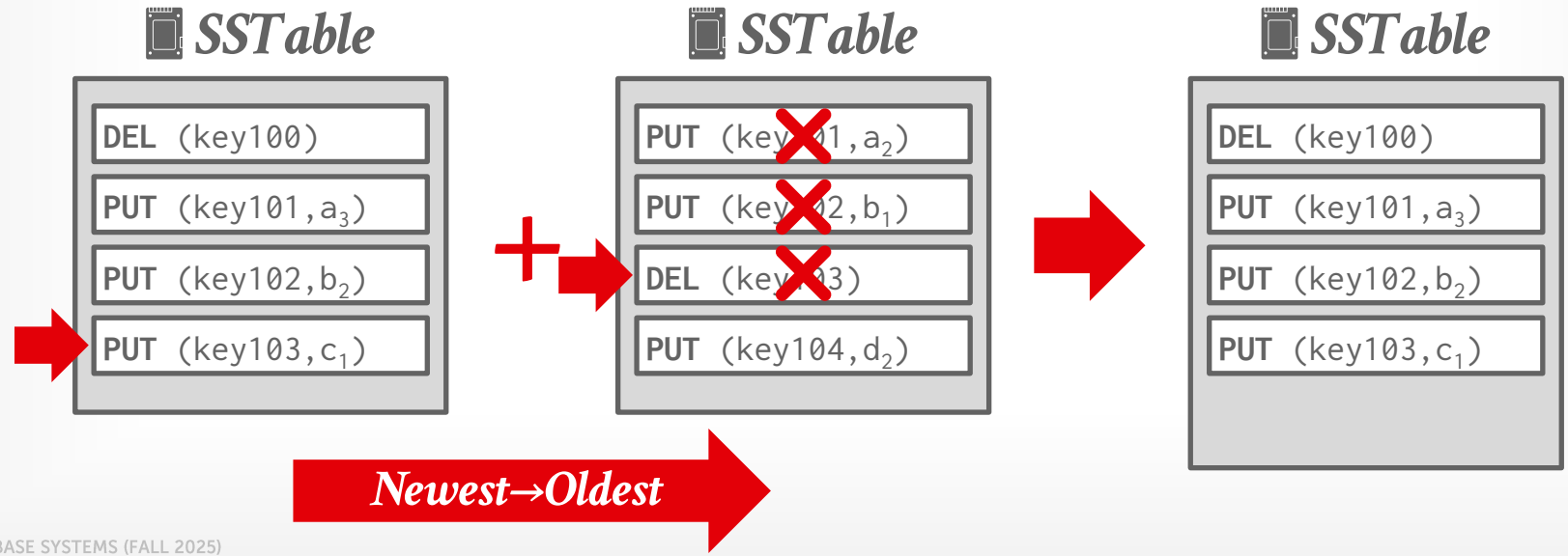

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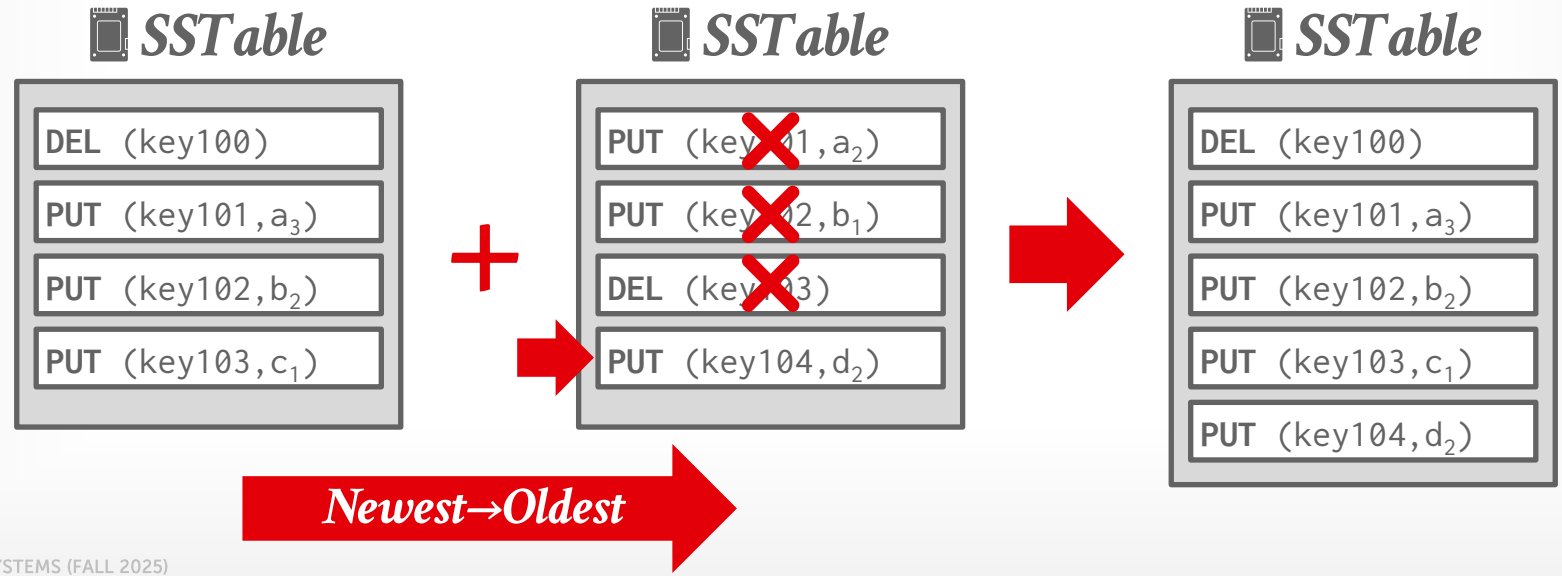

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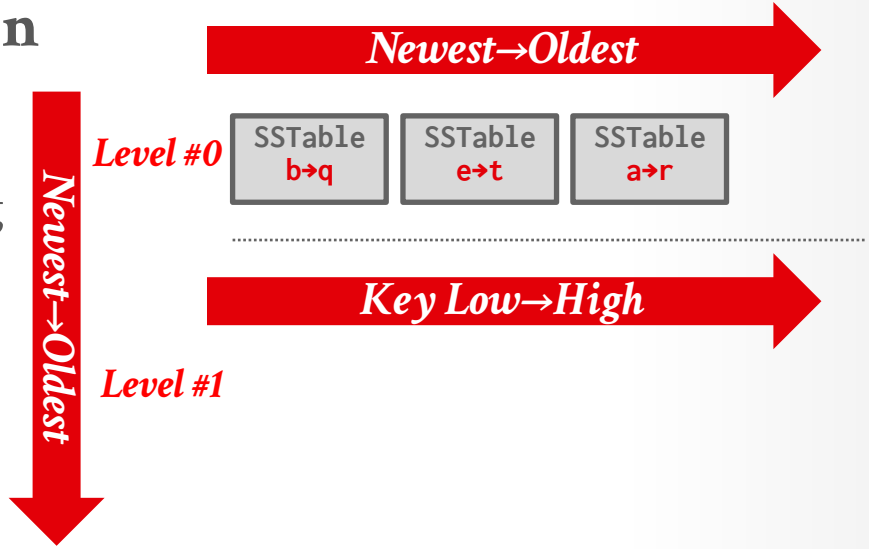

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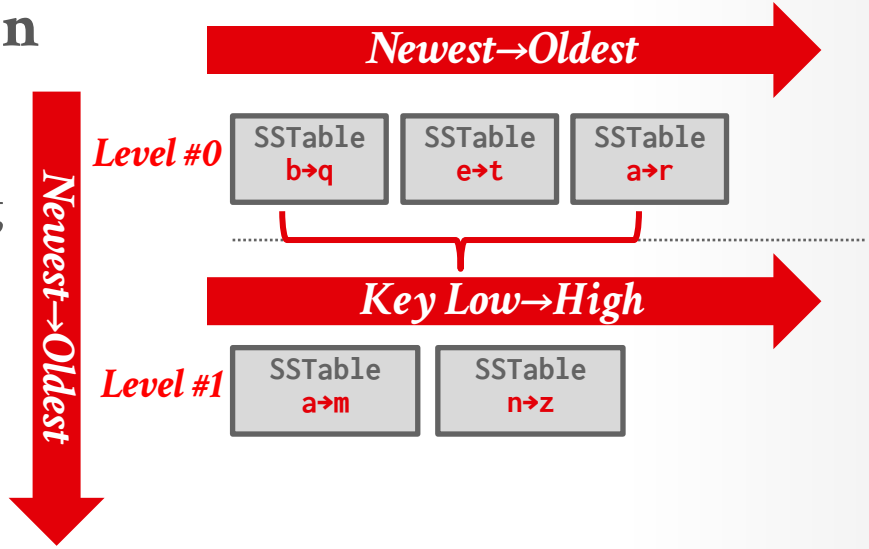

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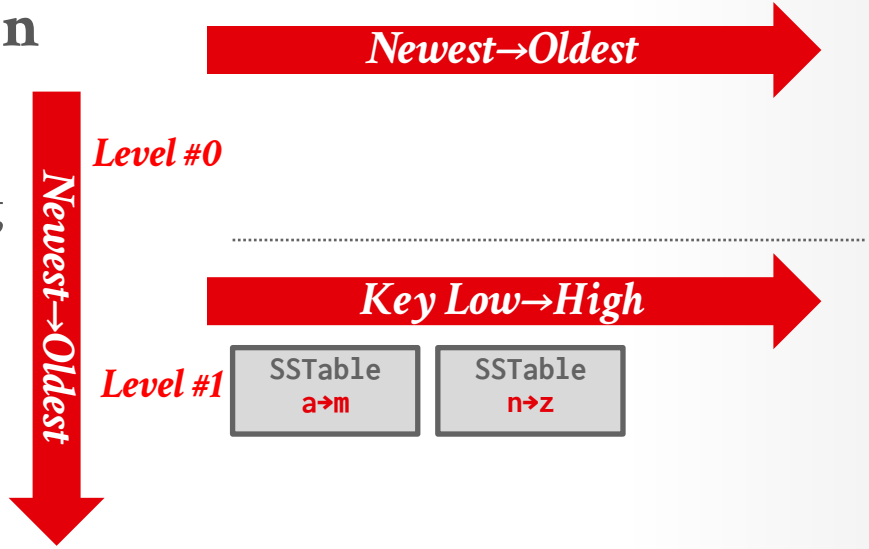

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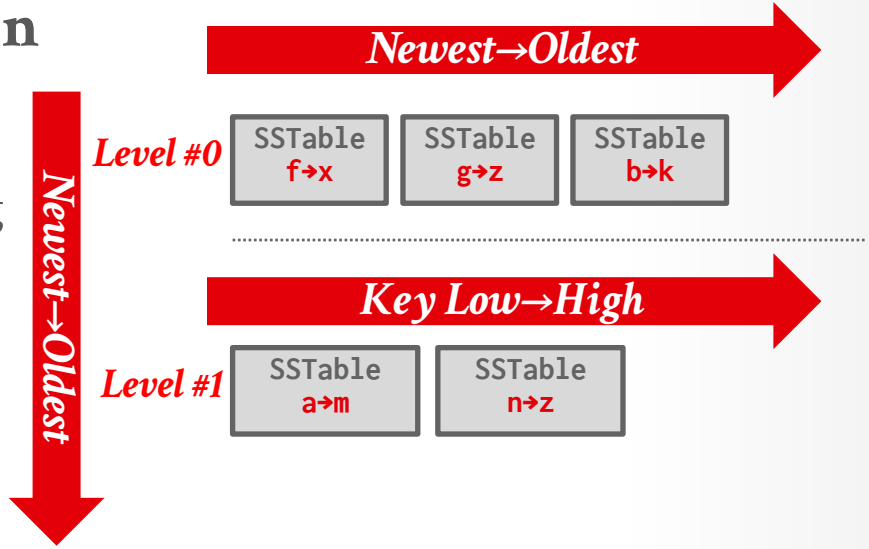

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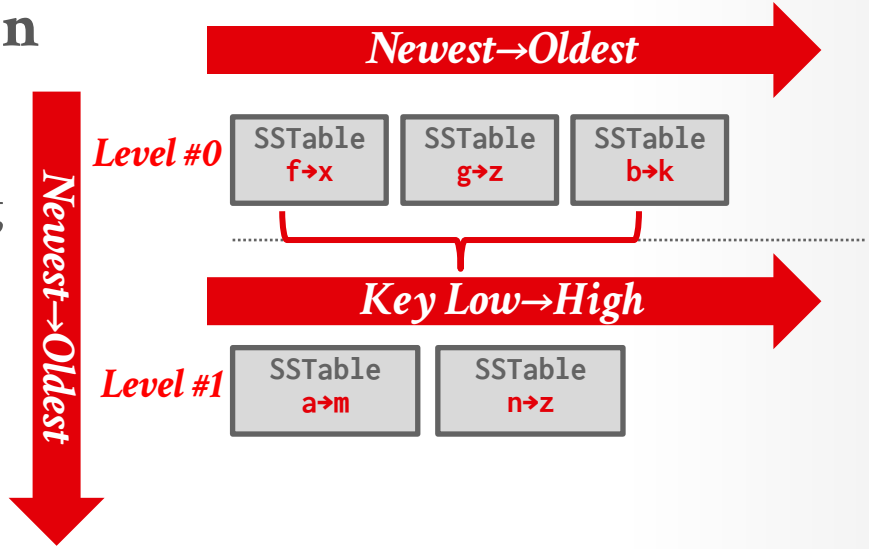

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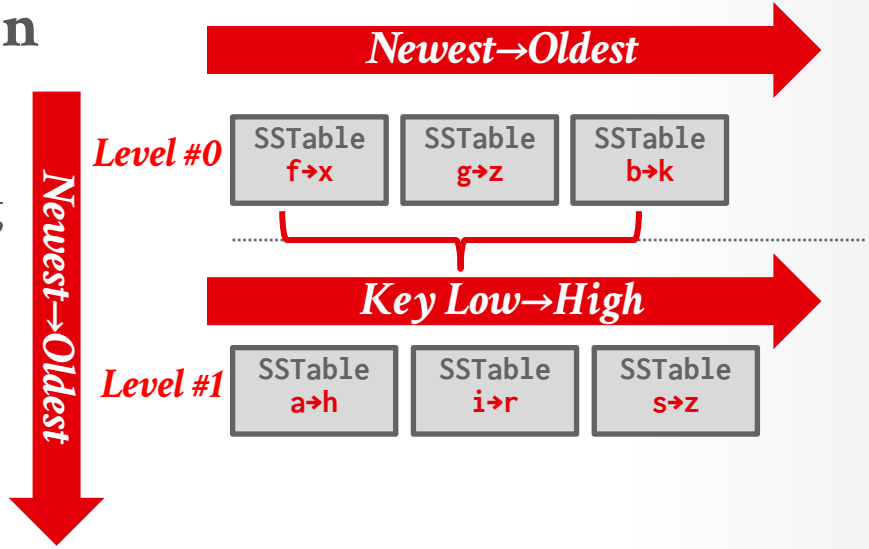

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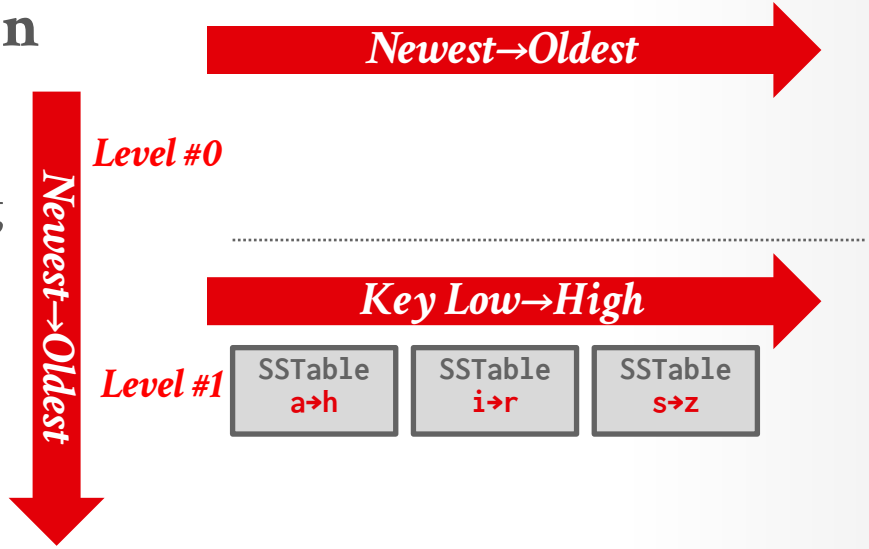

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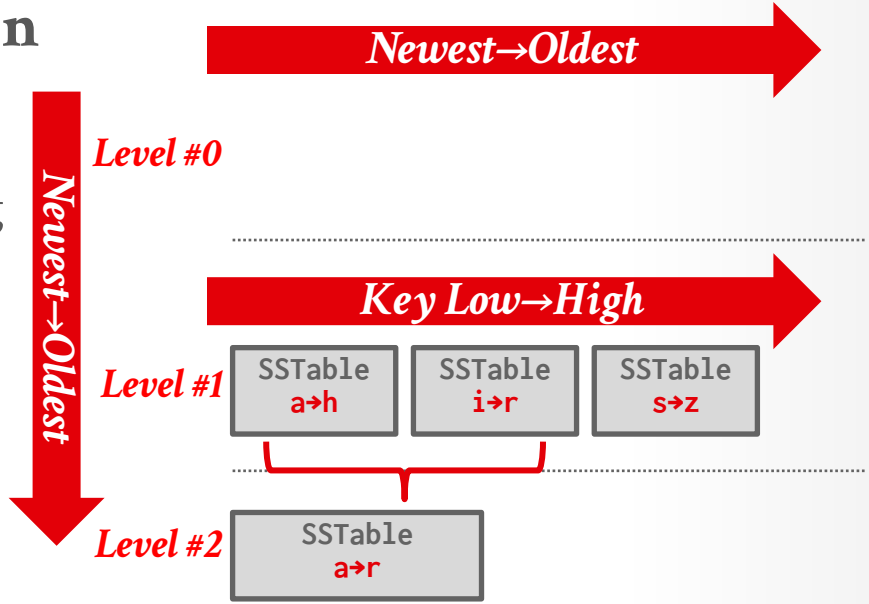

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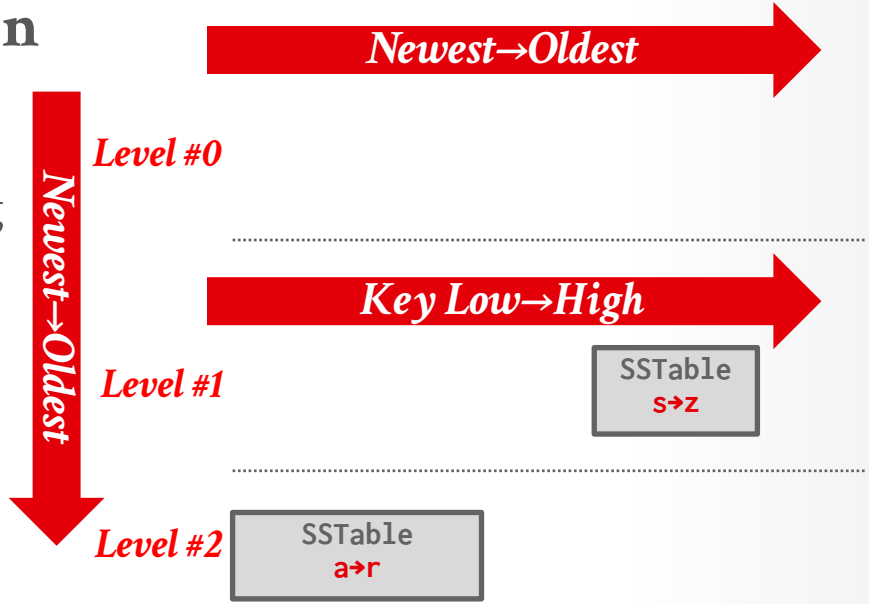

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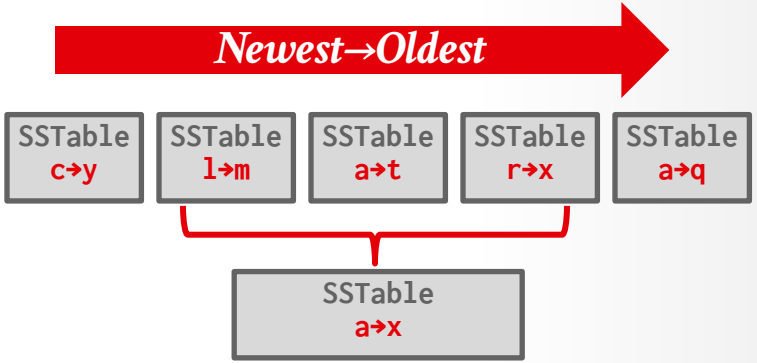

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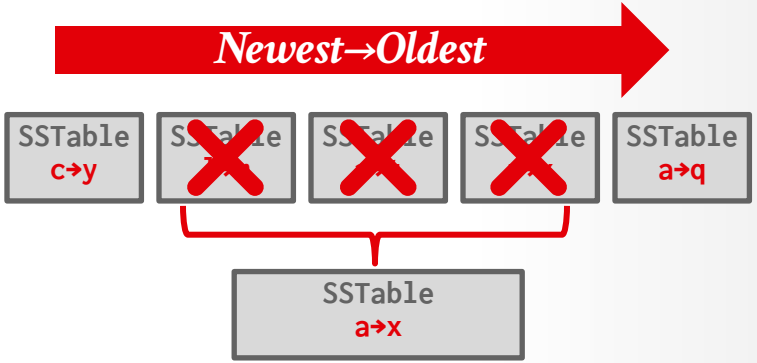

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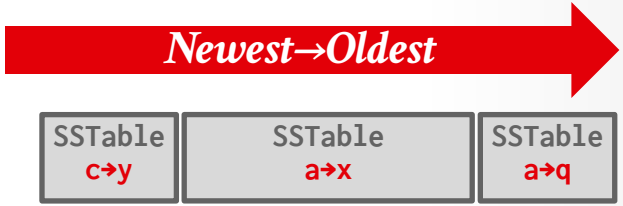

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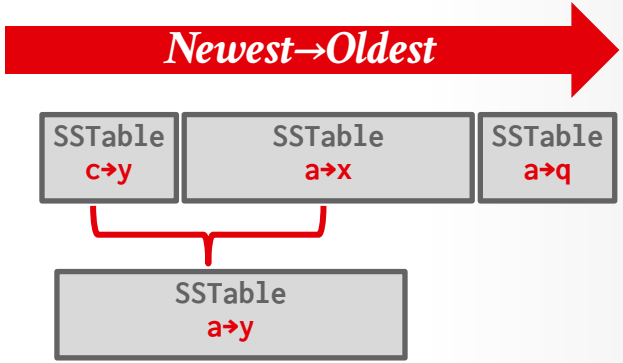

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DISCUSSION

Log-structured storage managers are more common today than in previous decades.
→ This is partly due to the proliferation of RocksDB.

What are some downsides of this approach?
→ Write-Amplification.
→ Compaction is expensive.

CONCLUSION

Log-structured storage is an alternative approach to the tuple-oriented architecture.

→ Ideal for write-heavy workloads because it maximizes sequential disk I/O.

The storage manager is not entirely independent from the rest of the DBMS.

NEXT CLASS

Breaking your preconceived notion that a DBMS stores everything as rows…